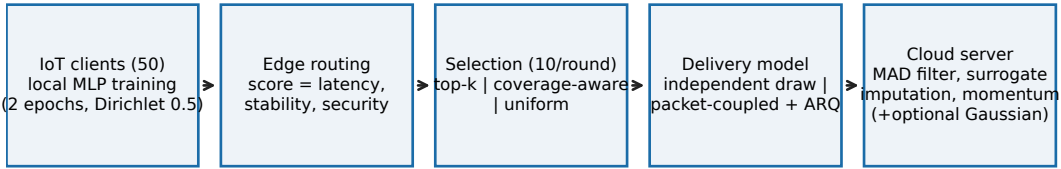
Simulated edge-cloud pipeline (one FL round); sign-flip attackers (20%) and client failures act on the update path



Evaluated protocol: validation-only; calibration -> threshold, selection -> checkpoint, report -> metrics. Final-test data locked.